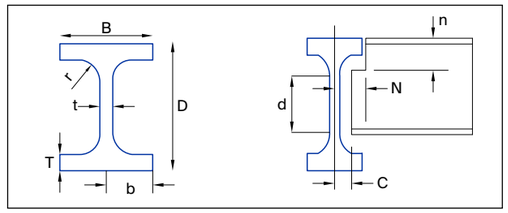
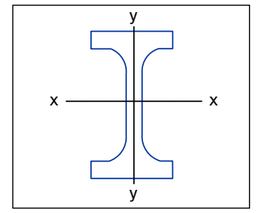
2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Imperial Units

Designation			Depth of section	Width of section	Thickness		Root radius	Depth between fillets	Area of section	Ratios for local buckling		Dimensions for detailing		
Size	Mass per metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W36-36x12 (914x305)	135	200.9	903.0	303.3	20.2	15.1	19.1	824.4	256	7.51	54.6	10	154	39
	150	224.2	910.4	304.1	23.9	15.9	19.1	824.4	286	6.36	51.8	10	154	43
	160	238.1	914.7	304.8	25.9	16.5	19.0	824.9	304	5.88	50.0	10	154	45
	170	253.4	918.4	305.5	27.9	17.3	19.1	824.4	323	5.47	47.7	11	154	47
	182	270.8	922.8	306.7	30.0	18.4	19.0	824.9	346	5.12	44.8	11	154	49
	194	289.1	926.6	307.7	32.0	19.5	19.1	824.4	368	4.81	42.3	12	154	51
	210	312.5	931.9	309.4	34.5	21.1	19.0	824.8	399	4.48	39.1	13	154	54
	232	345	942.9	307.8	39.9	22.1	19.1	824.9	440	3.86	37.3	13	153	59
	256	381	950.7	310.3	43.9	24.4	19.1	824.7	486	3.53	33.8	14	153	63
	286	426	960.9	312.8	48.0	26.9	19.1	826.7	536	3.26	30.7	15	153	67
	318	474	970.8	315.7	54.1	29.9	19.1	824.4	603	2.92	27.6	17	153	73
	350	521	980.7	318.8	58.9	33.0	19.1	824.7	663	2.71	25.0	19	153	78
	387	576	992.9	321.8	65.0	36.1	19.1	824.7	733	2.48	22.8	20	153	84
W36-36x16 1/2 (914x419)	230	343.3	911.8	418.5	32.0	19.4	24.1	799.6	437	6.54	41.2	12	210	56
	245	364.6	916.4	419.4	34.3	20.3	24.1	799.6	465	6.12	39.4	12	210	58
	260	388	921.0	420.5	36.6	21.4	24.1	799.6	494	5.74	37.4	13	210	61
	280	416.7	927.6	421.5	39.9	22.5	24.1	799.6	532	5.28	35.6	13	210	64
	300	446.4	933.2	423.0	42.7	24.0	24.1	799.7	569	4.96	33.3	14	210	67
	328	488.1	942.1	422.4	47.0	25.9	24.1	799.9	622	4.49	30.9	15	208	71
	359	534.2	950.0	424.9	51.1	28.5	24.1	799.7	680	4.16	28.1	16	208	75
	393	584.8	960.1	427.5	55.9	31.0	24.1	800.1	746	3.83	25.8	17	208	80
W40-40x12 (1016x305)	149	222	970.3	300.0	21.1	16.0	30.0	868.1	283	7.11	54.3	10	152	51
	167	249	980.2	300.0	26.0	16.5	30.0	868.2	317	5.77	52.6	10	152	56
	183	272	990.1	300.0	31.0	16.5	30.0	868.1	347	4.84	52.6	10	152	61
	211	314	1000.0	300.0	35.9	19.1	30.0	868.2	400	4.18	45.5	12	150	66
	235	349	1008.1	302.0	40.0	21.1	30.0	868.1	445	3.78	41.1	13	150	70
	264	393	1016.0	303.0	43.9	24.4	30.0	868.2	500	3.45	35.6	14	149	74
	294	438	1025.9	305.4	49.0	26.9	30.0	867.9	557	3.12	32.3	15	149	79
	327	487	1036.1	308.5	54.1	30.0	30.0	867.9	620	2.85	28.9	17	149	84
	359	535	1046.0	311.4	58.9	33.0	30.0	868.2	681	2.64	26.3	19	149	89

Surface area	Second moment of area		Radius of gyration		Elastic modulus		Plastic modulus		Buckling parameter	Torsional index	Warping constant	Torsional constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
2.96	325300	9424	35.7	6.07	7204	621	8351	982	0.854	46.8	18.4	291
2.97	376400	11240	36.3	6.27	8269	739	9535	1163	0.861	41.3	22.1	422
2.98	406100	12270	36.6	6.36	8879	805	10230	1266	0.864	38.6	24.2	515
2.99	436300	13300	36.8	6.42	9501	871	10940	1371	0.866	36.2	26.4	626
3.00	470700	14460	36.9	6.47	10200	943	11770	1487	0.866	33.9	28.8	767
3.01	504200	15600	37.0	6.51	10880	1014	12570	1601	0.867	31.9	31.2	926
3.03	549000	17130	37.1	6.55	11780	1107	13650	1754	0.866	29.7	34.5	1164
3.04	625200	19480	37.7	6.66	13260	1266	15340	2000	0.872	26.4	39.7	1648
3.06	696900	21980	37.9	6.72	14660	1416	17030	2247	0.872	24.1	45.2	2199
3.09	777000	24640	38.1	6.78	16170	1575	18870	2510	0.871	22.2	51.3	2891
3.11	884100	28580	38.3	6.89	18210	1810	21350	2895	0.872	19.9	60.0	4091
3.14	981300	32080	38.5	6.95	20010	2013	23590	3235	0.871	18.4	68.1	5337
3.17	110100	36460	38.8	7.05	22180	2266	26260	3654	0.871	16.8	78.5	7127
3.42	625800	39160	37.8	9.46	13730	1871	15480	2890	0.883	30.1	75.8	1193
3.43	671700	42240	38.0	9.53	14660	2014	16550	3111	0.884	28.3	82.2	1442
3.44	719600	45440	38.2	9.59	15630	2161	17670	3341	0.885	26.7	88.9	1734
3.45	785700	49870	38.4	9.68	16940	2366	19170	3658	0.887	24.8	98.3	2188
3.47	846900	53940	38.6	9.73	18150	2550	20600	3948	0.887	23.3	107	2671
3.48	936400	59170	38.8	9.76	19880	2801	22640	4344	0.887	21.4	119	3500
3.50	1031000	65450	38.9	9.81	21700	3081	24820	4790	0.886	19.8	132	4508
3.53	1144000	73000	39.2	9.89	23840	3415	27390	5320	0.887	18.2	149	5890
3.06	408000	9553	38.0	5.81	8409	637	9807	1020	0.850	45.7	21.5	390
3.08	481300	11760	39.0	6.09	9821	784	11350	1245	0.861	39.9	26.8	582
3.10	554000	14010	40.0	6.36	11190	934	12830	1470	0.873	35.0	32.2	835
3.11	644200	16240	40.1	6.37	12880	1083	14850	1713	0.872	30.7	37.7	1264
3.13	723100	18470	40.3	6.44	14350	1223	16590	1941	0.872	27.9	43.3	1718
3.14	807700	20500	40.2	6.40	15900	1353	18540	2168	0.868	25.5	48.4	2330
3.17	909900	23450	40.4	6.49	17740	1536	20760	2469	0.868	23.1	56.0	3185
3.19	1021000	26730	40.6	6.57	19720	1733	23200	2800	0.867	21.1	64.4	4299
3.22	1131000	29970	40.8	6.63	21620	1925	25570	3126	0.866	19.5	73.0	5576